

CS6765 NLP: Research Report

A Review of “GraphNarrator: Generating Textual Explanations for Graph Neural Networks”

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1 Summary

1.1 Motivation

Graph neural networks are widely used for tasks where the relations between entities matter, including molecular property prediction, financial fraud detection, community analysis in social networks, and recommendation systems. Although these models achieve impressive accuracy, they are notoriously opaque with regard to internal decision processes. In most cases, users only have the final prediction, without any understanding of how different parts of the graph influenced the result. Lack of transparency might be a limiting factor for trust, debugging, and application in sensitive domains where justification is necessary.

Most previous approaches for explaining GNNs rely on saliency-based visualizations, highlighting important nodes or edges. While useful for expert users, these methods remain unintuitive to general users. Visual or structural explanations do not convey full reasoning and often require domain expertise to interpret correctly. The gap from low-level graph saliency to high-level human reasoning creates a strong motivation for explanations that “speak the user’s language”.

While promising, natural language explanations are far from trivial to construct from GNNs. A fundamental challenge arises when translating non-linear, non-sequential, and topology-driven representations of structural graph information into coherent language that truly reflects the GNN’s actual reasoning. Most previous LLM-based explanations suffer from either hallucination, over-generalization, or a lack of grounding. Motivated by these aspects, the authors advance GraphNarrator—a framework that produces explanations both accurate and genuinely reflective of GNN decision-making.

1.2 Contribution

This paper is about GraphNarrator, a system that helps explain graph predictions using language we can understand. It mixes most important improving the explanation bit by bit, and shrinking down a bigger model to make things faster. Its main contributions are:

- A structured transformation pipeline to convert raw graph saliency signals into “saliency paragraphs”—clear textual summaries that describe which graph substructures influence a prediction and why.
- An expert-iteration refinement framework that assesses candidate explanations based on criteria designed specifically for GNN interpretability: input faithfulness, output alignment, and conciseness.
- A distillation-based explainer model that learns to produce final explanations in one step. This greatly improves inference speed and makes the method deployable in real-world scenarios.
- Conducting a comprehensive evaluation with both automatic metrics and human assessments on three Tree annotated Graph datasets exhibit an improvement in factual grounding, readability, and human preference over existing baselines such as GNNExplainer and LLM prompting methods.

Overall, the paper contributes a unique and effective approach to symbolic reasoning. Graphs supported by textual explanation techniques from NLP

1.3 Methodology

The GraphNarrator framework consists of three major stages. First, the system extracts saliency information from a trained GNN to determine which nodes and edges are most influential for the prediction. Traditional saliency maps are hard to interpret by end-users; hence, the authors verbalize them into saliency paragraphs. These paragraphs describe important substructures, their contextual roles, and their contribution to the model’s decision. Second, the expert-iteration process refines these explanations. Within each iteration, an LLM produces an explanation which is subsequently assessed by a critic model over three dimensions. Input faithfulness guarantees the interpretation is representative of the structural evidence found by the saliency method. Output faithfulness ensures the explanation stays aligned with the predicted class instead of inventing new justifications. Brevity encourages concise communication, reducing redundancy and improving readability.

The LLM receives feedback and generates updated explanations, gradually improving clarity and grounding. This iterative approach resembles reinforcement learning or human guided critique loops commonly used in modern NLP, but adapted specifically for graph explanations.

In the final stage, the entire multi-step pipeline is distilled into a single explainer model. The distillation phase trains the LLM to imitate the high-quality explanations produced during expert iteration, enabling one-step inference. This greatly reduces computational cost and removes the need for repeated refinement during deployment.

The authors argue that this combination grounding via saliency, refining via expert feedback, and simplifying via distillation provides a robust methodology for generating faithful, human readable explanations for GNNs.

1.4 Conclusion

The evaluation shows that GraphNarrator gives more accurate, concise, and understandable explanations compared to the state-of-the-art graph explanation methods. Automatic metrics indicate

better alignment of the explanations with the model’s saliency signals. Human evaluators also prefer GraphNarrator’s explanations for clarity, relevance, and usefulness.

The authors have underlined that natural language explanations can definitely improve user trust in GNN-based systems, especially when the domain requires strong justification. They further indicate limitations: reliance on LLMs increases computational cost, and experiments are restricted to moderately sized datasets. GraphNarrator represents a solid basis for integrating natural language reasoning with graph-based learning models and opens several directions for further investigation.

2 Critique

2.1 Strengths

One of the high points of the paper is the clear motivation and practical relevance of the problem it addresses. Explainability becomes an increasingly crucial issue while machine learning models are being fielded in sensitive areas such as health care, legal systems, and financial analysis. In this respect, the fact that GNNs are a model family already demonstrating real world impact makes the author’s work timely and relevant.

Another strength is the utilization of a structured iterative refinement process. This work avoids either ad hoc prompting or single pass LLM generation; instead, the authors systematically apply criteria to evaluate and improve explanations. This methodology differentiates GraphNarrator from the prior work and goes a long way toward helping ensure that the generated text is grounded, concise, and aligned with the model’s internal reasoning.

Another particularly salient aspect is the distillation step. Many explanation techniques are hard to deploy in practice, since they require multiple passes or complicated training pipelines. The mere fact that the authors compress the whole iterative process into a single efficient model speaks volumes about attention to usability and scalability, qualities lacking in many academic explanation systems.

2.2 Weaknesses and Limitations

The main limitation of the approach is its dependence on the quality of saliency maps. Saliency methods are known to be noisy and sometimes inconsistent for GNNs. If the saliency extraction itself is inaccurate, even a well-phrased explanation will mislead one. Although the process of expert iteration does help alleviate this, it cannot fully compensate for an unreliable foundation.

Another limitation is that this approach assumes that explanations can be fully captured by textual descriptions. Some graph predictions rely on subtle structural interactions that are inherently hard to phrase in natural language. For such predictions, textual explanations will oversimplify the model’s reasoning. A combination of structured diagrams and text may work better.

Finally, the reliance on LLMs raises concerns about reproducibility and computational accessibility. Training and running LLMs—even distilled ones—is expensive, and this may inhibit the adoption of GraphNarrator in resource-constrained settings.

3 Synthesis

GraphNarrator’s ideas connect strongly with course topics such as interpretability, natural language generation, alignment, and multimodal reasoning. In particular, it illustrates how explanation principles traditionally associated with NLP—such as rationale extraction or chain-of-thought—can be extended to non-text domains. The method reflects a broader trend in machine learning: combining symbolic or structured signals with the expressive power of LLMs.

The framework also aligns with discussions on model alignment. The expert-iteration loop strongly resembles reinforcement learning from human feedback (RLHF) and preference optimization techniques used to align LLMs with human expectations. An interesting twist on standard alignment techniques, GraphNarrator aligns the explanations to model behavior and structural evidence instead of directly aligning them to human judgment.

From an NLP perspective, this work raises important questions about the way language can serve as a medium for communicating complex model behavior. Similar to explanations in sentiment classification, question answering, or summarization, GraphNarrator demonstrates that natural language can act as an interpretability layer even when the underlying data is not linguistic. One avenue is the extension of the method to larger and more diverse graph datasets, such as molecular graphs, knowledge graphs, and financial transaction networks. Another promising direction is the integration of domain knowledge or symbolic reasoning into the explanation process to enhance factual accuracy. Lastly, methods from this work could be adapted for multimodal explanation systems where text, images, and structured data come together to form richer explanations. Overall, GraphNarrator is a great example of how NLP techniques more broadly, LLM-based reasoning—can be used to improve interpretability across different areas of machine learning.

References

- [1] Bo Pan, Zhen Xiong, Guanchen Wu, Zheng Zhang, Yuntong Hu, Yifei Zhang, and Liang Zhao. *GraphNarrator: Generating Textual Explanations for Graph Neural Networks*. Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (ACL), 2025.